

MATHEMATICAL METHODS UNIT 3

QUIZ 3B (Tech-Active)

Test theme: Polynomials, Exponential and Logarithmic Functions

SOLUTIONS

SECTION A – Multiple-choice questions

Answers

Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10
C	D	D	E	C	B	D	C	C	E

Question 1

The exact value of the x -intercept of $y = 7 \log_2(1 - 6x) + 9$ is

- A. (0.121, 0)
 B. (0.098, 0)
 C. $\left(\frac{1}{6} - \frac{1}{12 \times 2^{\frac{2}{7}}}, 0\right)$
 D. $\left(\frac{1}{6} - \frac{1}{6 \times e^{\frac{9}{7}}}, 0\right)$
 E. (9, 0)

y-intercept: Substitute $x = 0$. $7 \log_2(1 - 6x) = 0 \Rightarrow x = \frac{1}{6} - \frac{1}{12 \times 2^{\frac{2}{7}}}$ Solve[7 Log[2, 1 - 6 x] + 9 == 0, x] // Expand $\left\{ \left\{ x \rightarrow \frac{1}{6} - \frac{1}{12 \times 2^{2/7}} \right\} \right\}$	C
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Question 2

$\log_e(5ab)^2 + \log_e(ba)$ expressed as a single logarithm if possible is

- A. $\log_e(5ab)^2 + \log_e(ba)$ cannot be expressed as a single logarithm
 B. $2 \log_e(5a^2b^2)$
 C. $\log_e(25ab)$
 D. $\log_e(25a^3b^3)$
 E. $\log_e(10a^3b^3)$

$\log_e(25a^2b^2) + \log_e(ba) = \log_e(25a^3b^3)$ (←best to be done by hand←) Log[5 a b]^2 + Log[b a] // Simplify Log[a b] + Log[5 a b]^2 (←watch out separate the a and b or mathematica thinks it is one variable←) Log[5 a b]^2 + Log[b a] // Simplify Log[5 a b]^2 + Log[b a]	D
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Question 3

The function $f : [a, \infty) \rightarrow \mathbb{R}$ with rule

$f(x) = \log_e(x^4)$ will have an inverse function if

- A. $a < 0$
- B. $a \leq -1$
- C. $a \leq 1$
- D. $a > 0$
- E. $a \geq -1$

Question 4

The function g has the rule $g(x) = \log_e(x - b)$, where b is a real constant.

The maximal domain of g is

- A. $\mathbb{R}^+$
- B. $\mathbb{R} \setminus \{b\}$
- C. $\mathbb{R}$
- D. $(-b, b)$
- E. (b, ∞)

Question 5

The equation $f(x) = 0$ is to be solved using the bisection method.

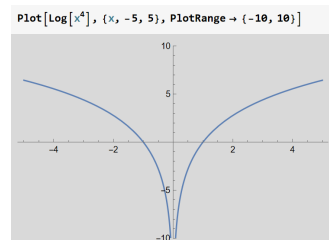
If $f(3) = 4$ and $y = f(x)$ has one vertical asymptote with equation $x = -4$, which one of the following is an appropriate choice for an endpoint of the initial interval?

- A. -5 such that $f(-5) = -26$
- B. 0 such that $f(0) = 5$
- C. 1 such that $f(1) = -2$
- D. 13 such that $f(13) = 2$
- E. 3 such that $f'(3) = 0$

For the inverse function to exist the function must be one-to-one.

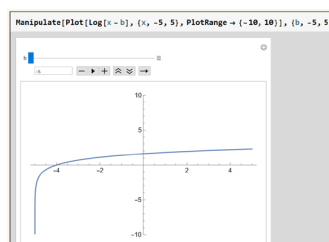
Therefore the graph of the function needs to pass the horizontal line test: A horizontal line must not cut the graph more than once.

The domain $[a, \infty)$ implies 'retain the right-hand half' of the graph.



D

$x - b > 0$ is required: $x > b$.



E

In the bisection method, a solution to $f(x) = 0$ exists in the interval $[a, b]$ only if $f(a) \times f(b) < 0$ and f is continuous over the interval.

The requirement $f(a) \times f(b) < 0$ eliminates options B, D and E.

The requirement for continuity eliminates option A because there is a discontinuity (asymptote at $x = -4$) in the interval $[-5, 3]$.

C

Question 6

The graph of $y = kx - 3$ intersects the graph of $y = -x^2 + 2x - 12$ at two distinct points for

- A. $k \in (-\infty, -8) \cup (4, \infty)$
 B. $k \in (-\infty, -4) \cup (8, \infty)$
 C. $k \in (-4, 8)$
 D. $k \in [-4, -8]$
 E. $k \in [-4, 8]$

Solve $y = kx - 3$ and $y = -x^2 + 2x - 12$ simultaneously:

B

$$kx - 3 = -x^2 + 2x - 12$$

$$\Rightarrow 0 = -x^2 + 2x - 12 - (kx - 3)$$

$$\Rightarrow -x^2 + (2 - k)x - 9 = 0.$$

For two solutions:

$$\Delta = b^2 - 4ac > 0 \quad \Rightarrow (2 - k)^2 - 4(-1)(-9) > 0$$

therefore $k \in (-\infty - 4) \cup (4, \infty)$.

Reduce [Discriminant $[-x^2 + 2x - 12 - (kx - 3), x] > 0, k]$

$k < -4 \mid k > 8$

Question 7

Which one of the following is a list of all possible rational roots of the polynomial $3x^4 + 2x^3 - 28x^2 - 18x + 9$?

- A. $\frac{1}{1}, \frac{1}{3}, \frac{3}{1}, \frac{3}{3}, \frac{9}{1}, \frac{9}{3}$
 B. $3, -3, -1, \frac{1}{3}$
 C. $\pm 1, \pm 3, \pm 9, \pm \frac{1}{3}, \pm \frac{3}{3}, \pm \frac{9}{3}$
 D. $\pm 1, \pm 3, \pm 9, \pm \frac{1}{3}$
 E. $\pm 1, \pm 3, \pm \frac{1}{3}, \pm \frac{1}{9}$

$$\begin{aligned} \text{Possible rational roots} &= \frac{\text{factors of constant term}}{\text{factors of leading term}} \\ &= \frac{\pm 1, \pm 3, \pm 9}{\pm 1, \pm 3} \\ &= \{\pm 1, \pm \frac{1}{3}, \pm 3, \pm 9\} \end{aligned}$$

Equivalent fractions must be removed.

D

Question 8

Let the function $f: R \rightarrow R$, $f(x) = e^x + e^{-x}$ for all $u \in R$.

$f(2u)$ is equivalent to

- A. $f(u) + f(-u)$
 B. $2f(u)$
 C. $(f(u))^2 - 2$
 D. $(f(u))^2$
 E. $(f(u))^2 + 2$

$$\begin{aligned} f(2u) &= e^{2u} + e^{-2u} \\ (f(u))^2 &= (e^u + e^{-u})^2 \\ &= (e^u + e^{-u})(e^u + e^{-u}) \\ &= e^{2u} + 2e^u e^{-u} + e^{-2u} \\ &= e^{2u} + 2e^{u-u} + e^{-2u} \\ &= e^{2u} + 2e^0 + e^{-2u} \\ &= e^{2u} + e^{-2u} + 2 \\ &= f(2u) + 2 \\ \therefore f(2u) &= (f(u))^2 - 2 \end{aligned}$$

f[x_] := Exp[x] + Exp[-x]

f[2u]

$e^{-2u} + e^{2u}$

(f[u])^2 - 2 // Expand

$e^{-2u} + e^{2u}$

C

Question 9

If $\log_5(2.5) = p$, then $\log_5(2)$ is equal to

- A. $p - 0.5$
 B. $p - \log_5(0.5)$
 C. $1 - p$
 D. $p - 1$
 E. $1.25p$

METHOD 1: Use log laws. $\log_5(2.5) = p$ $\log_5\left(\frac{5}{2}\right) = p$ $\log_5(5) - \log_5(2) = p$ $1 - \log_5(2) = p$ $\log_5(2) = 1 - p$	METHOD 2: Guess and check using <i>Mathematica</i> . Log[5, 2.5] 0.569323 1 - Log[5, 2] // N 0.569323	A
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Question 10

The graph of a polynomial function with rule $y = f(x)$ has a local maximum at the point $(a, f(a))$.

The graph also has a local minimum at $(0, 0)$ but no other stationary points.

The graph of the function with the rule $y = -2f\left(\frac{x}{2}\right) + k$, where k is a positive real number, has

- A. a local maximum at the point with coordinates $(2a, -2f(a) + k)$
 B. a local maximum at the point with coordinates $\left(\frac{a}{2}, -2f(a) + k\right)$
 C. a local maximum at the point with coordinates $(2a, -2f(a) - k)$
 D. a local minimum at the point with coordinates $\left(\frac{a}{2}, 2f(a) + k\right)$
 E. a local minimum at the point with coordinates $(2a, -2f(a) + k)$

Maximum turning point at $(a, f(a))$. A correct sequence of transformations mapping $y = f(x)$ to $y = -2f\left(\frac{x}{2}\right) + k$ must be used: 1. Reflect in the x-axis (affects y-value), max TP becomes a min TP $(a, -f(a))$. 2. Dilate factor 2 from x-axis (affects y-value) $(a, -2f(a))$. 3. Dilate factor 2 from the y-axis (affects x-value) $(2a, -2f(a))$. 4. Up k units (affects y-value) $(2a, -2f(a) + k)$.	E
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END OF SECTION A

SECTION B

Question 1 (20 marks)

Queenie the chicken is walking around the backyard. She is on a path that she regularly follows. Ada thinks this is very interesting behaviour and decides to map Queenie's favourite path using a quartic equation. Queenie's path is defined by the function

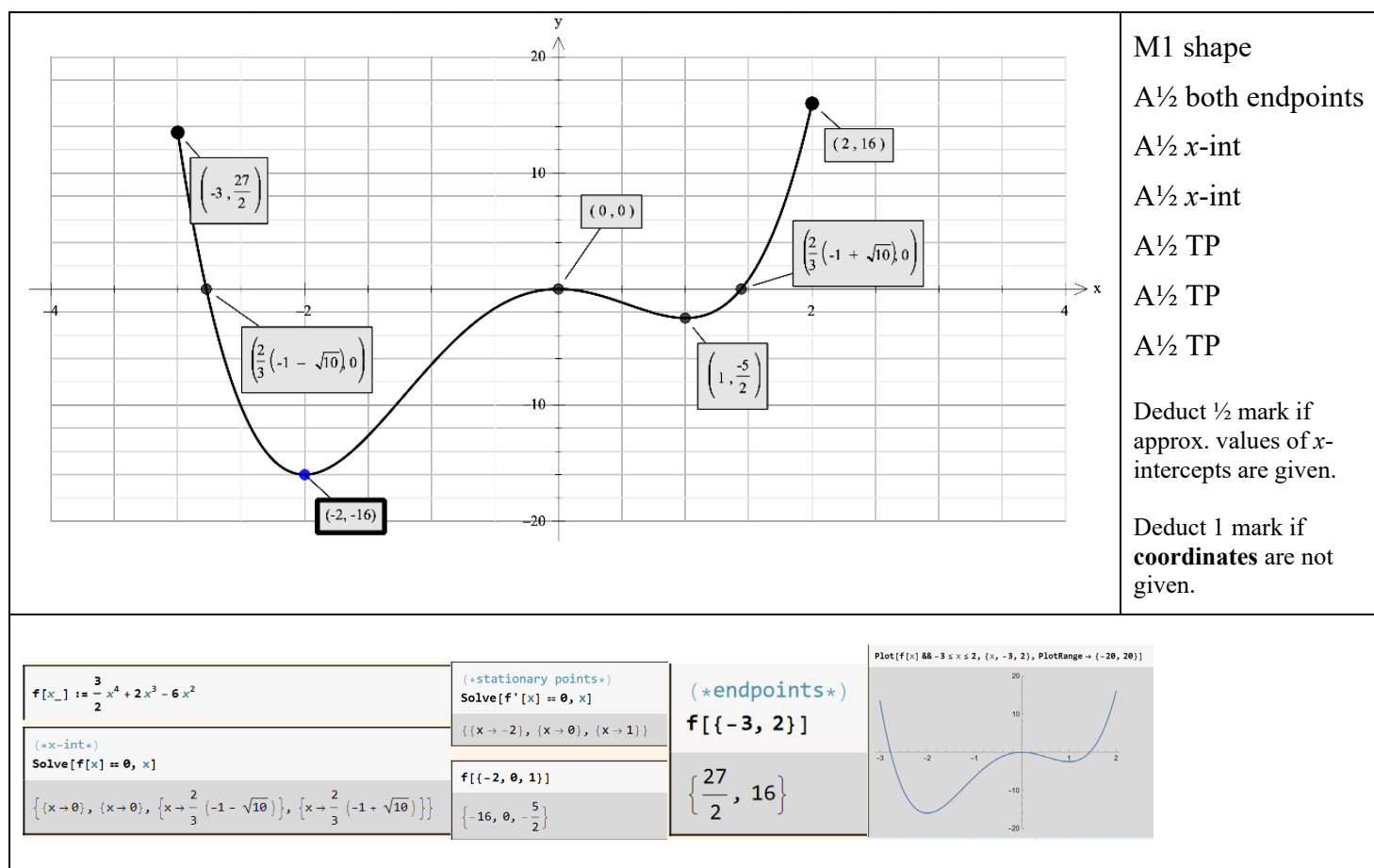
$$f: [-3, 2] \rightarrow \mathbb{R}, f(x) = \frac{3}{2}x^4 + 2x^3 - 6x^2,$$

where the x -axis represents east/west direction and the y -axis represents the north/south direction. All measurements are in metres.

- a. Sketch a graph of $y = f(x)$ on the axes below.

Label any axes intercepts, stationary points and endpoints with their coordinates.

4 marks



- b. State the values of $b \in \mathbb{R}$ for which the graph of $y = f(x) + b$ has no x -intercepts

1 mark

Answer: $b \in (-\infty, -16) \cup (16, \infty)$.

Solution:

1. Consider the absolute minimum turning point $(-2, -16)$.

Translate the graph 16 units up and there will be no x -intercepts.

2. Consider the absolute maximum endpoint $(2, 16)$.

Translate the graph 16 units down and there will be no x -intercepts.

A1

Ada wants to help Queenie and decides to put some chicken seed on the x -intercepts of her favourite path.

- c. Ada decides to estimate an x -intercept by using the bisection method. She writes the following pseudocode for estimating a solution to $f(x) = 0$ using the bisection method.

Inputs: $f(x)$, a function of x , where x is a real number
 a , the lower interval endpoint
 b , the upper interval endpoint
 max , the maximum number of iterations

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Define bisection( $f(x)$ ,  $a, b, \text{max}$ )
  If  $f(a) \times f(b) > 0$  Then
    Return "Invalid interval"
   $i \leftarrow 1$ 
  While  $i < \text{max}$  Do
     $\text{mid} \leftarrow (a + b) \div 2$ 
    If  $f(\text{mid}) = 0$  Then
      Return  $\text{mid}$ 
    Else If  $f(a) \times f(\text{mid}) < 0$  Then
       $b \leftarrow \text{mid}$ 
    Else
       $a \leftarrow \text{mid}$ 
     $i \leftarrow i + 1$ 
  EndWhile
  Return  $\text{mid}$ 

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The algorithm is implemented as follows:

`bisection((3/2)x^4 + 2x^3 - 6x^2, -3, -1, 4)`

- i. What is the initial interval?

1 mark

Answer: $[a, b] = [-3, -1]$	A1
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- ii. What value, correct to two decimal places, is returned when the above algorithm is implemented?

1 mark

Answer: -2.75. Solution: First iteration $i = 1$, second iteration $i = 2$, third iteration $i = 3$. The algorithm then stops as the maximum number of iterations is less than 4.	A1
Input: <pre> Clear[a,b,f,nit] f[x_]:=2/3 x^4+2 x^3-6 x^2 (* Insert the function f(x) from the equation f(x)=0 to be solved *) a[1]=-3 b[1]=-1 (* Insert the given interval (a, b) *) (* Note: f(a) and f(b) must have opposite signs: f(a)*f(b) < 0 *) If[Sign[f[a1]]*Sign[f[b1]]>0, Print[Style["All OK, execute the algorithm.",Blue]], Print[Style["f(a) and f(b) do not have opposite signs. Choose different values of a and b.",Red]] (* Check that f(a)*f(b) < 0 *) nit=3 (* Number of iterations *) All OK, execute the algorithm. </pre>	Algorithm: <pre> Do[x[n]=(a[n]+b[n])/2; (* x[n] represents x_n, x[1] is the value of x_1 *) If[f[x[n]]*f[b[n]]<0, a[n+1]=x[n];b[n+1]=b[n], a[n+1]=a[n];b[n+1]=x[n]],{n,nit}]; Style[Row[{"Approximate solution to ",f[x],"= 0 after ",nit," iterations is ",N[x[nit],6]}],FontColor->Blue] (* Print out approximate solution *) </pre> Approximate solution to $-6x^2 + 2x^3 + \frac{3x^4}{2} = 0$ after 3 iterations is -2.75000



d. Astro walks in a straight line across the yard and meets Queenie's path, $y = f(x)$, at $x = -\frac{1}{3}$.

1 mark

Answer: $-\frac{13}{18}$.	$f\left[-\frac{1}{3}\right]$ $-\frac{13}{18}$	A1
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2 marks

$m = f' \left(-\frac{1}{3} \right) = \frac{40}{9}.$ Substitute $m = \frac{40}{9}$ and the point $\left(-\frac{1}{3}, -\frac{13}{18} \right)$ from part d. i. Method 1: Use the model $y = mx + c$. $-\frac{13}{18} = \frac{40}{9} \left(-\frac{1}{3} \right) + c \quad \Rightarrow c = \frac{41}{54}.$ $y = \frac{40}{9}x + \frac{41}{54}.$ Rearrange into the required form: $y = \frac{240}{54}x + \frac{41}{54} \quad \Rightarrow 54y = 240x + 41$ $\Rightarrow 54y - 240x - 41 = 0.$ Method 2: Use the model $y - y_1 = m(x - x_1)$.	$f' \left[-\frac{1}{3} \right]$ $\frac{40}{9}$ $\text{Solve} \left[f \left[-\frac{1}{3} \right] = f' \left[-\frac{1}{3} \right] * -\frac{1}{3} + c, c \right]$ $\left\{ \left\{ c \rightarrow \frac{41}{54} \right\} \right\}$	M1: Find m and substitute the point from part d.i.
		A1: Answer in correct form with full working shown.

- e. i. The straight line Astro walks in intersects $y = f(x)$ at the point where $x = -\frac{1}{3}$ and at two other points. State the x -coordinates of the two other points of intersection. Express your answers in the form $\frac{a \pm \sqrt{b}}{c}$ where a , b , and c are integers. 2 marks

Solve $\frac{3}{2}x^4 + 2x^3 - 6x^2 = \frac{40}{9}x + \frac{41}{54}$: $\underbrace{\frac{3}{2}x^4 + 2x^3 - 6x^2}_{\text{Queenie's path } y=f(x)}$ $\underbrace{\frac{40}{9}x + \frac{41}{54}}_{\text{Astro's path}}$ Answer: $x = \frac{-1 \pm \sqrt{42}}{3}$.	Solve[$f[x] == \frac{40}{9}x + \frac{41}{54}, x]$ $\{\{x \rightarrow -\frac{1}{3}\}, \{x \rightarrow -\frac{1}{3}\}, \{x \rightarrow \frac{1}{3}(-1 - \sqrt{42})\}, \{x \rightarrow \frac{1}{3}(-1 + \sqrt{42})\}\}$	M1: Two equations equal to each other. A1: Correct answers in correct form.
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- ii. Show that the distances between $x = -\frac{1}{3}$ and each intersection point found in **part e. i.** are equal, correct to two decimal places. 2 marks

Distance between $\left(-\frac{1}{3}, -\frac{13}{18}\right)$ and $\left(\frac{-1 + \sqrt{42}}{3}, \frac{1}{54}(-39 + 80\sqrt{42})\right)$: 9.84 meters Distance between $\left(-\frac{1}{3}, -\frac{13}{18}\right)$ and $\left(\frac{-1 - \sqrt{42}}{3}, \frac{1}{54}(-39 - 80\sqrt{42})\right)$: 9.84 metres Therefore distance is the same for both sections of Astro's path.	$f\left[\frac{1}{3}(-1 + \sqrt{42})\right] // \text{Simplify}$ $\frac{1}{54}(-39 + 80\sqrt{42})$ numerical value integer part mini $f\left[\frac{1}{3}(-1 - \sqrt{42})\right] // \text{Simplify}$ $\frac{1}{54}(-39 - 80\sqrt{42})$ EuclideanDistance[$\{(-\frac{1}{3}, -\frac{13}{18}), (\frac{1}{3}(-1 + \sqrt{42}), \frac{1}{54}(-39 + 80\sqrt{42}))\}$] // N 9.84112 EuclideanDistance[$\{(-\frac{1}{3}, -\frac{13}{18}), (\frac{1}{3}(-1 - \sqrt{42}), \frac{1}{54}(-39 - 80\sqrt{42}))\}$] // N 9.84112	M1: Two coordinates found using their x-coordinates from part e.i. H1: Distances confirmed correct to 2 dp.
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Ada is not wanting to return to work and procrastinates further about Queenie's favourite path. Though, she likes to think she is practising mindfulness ... which of course she is!

Ada decides to manipulate the quartic equation with unknown parameters to change the features of the path.



Let the function $p : R \rightarrow R$, $p(x) = \frac{3}{2}x^4 + 2x^3 + 3(a-2)x^2 - 6ax + \frac{1}{2}a^2$ where a is a real constant.

f. State the value of a for which $f(x) = p(x)$ for all x .

1 mark

Answer: $a = 0$.

Solution:

Require $f(x) = \frac{3}{2}x^4 + 2x^3 - 6x^2$ to be the same as

$$p(x) = \frac{3}{2}x^4 + 2x^3 + 3(a-2)x^2 - 6ax + \frac{1}{2}a^2.$$

Method 1: Equate coefficients.

$$\text{Equate constant terms: } 0 = \frac{1}{2}a^2 \Rightarrow a = 0.$$

Check: Equate coefficients of x or x^2 .

METHOD 2:

$$\text{SolveAlways}\left[\frac{3}{2}x^4 + 2x^3 - 6x^2 == \frac{3}{2}x^4 + 2x^3 + 3(a-2)x^2 - 6ax + \frac{1}{2}a^2, x\right]$$

$$\{\{a \rightarrow 0\}\}$$

A1

g. Find, in terms of a where appropriate, all solutions to $p'(x) = 0$.

1 mark

Answer: $x = 1, -1 \pm \sqrt{1-a}$.

$$p[x_] := \frac{3}{2}x^4 + 2x^3 + 3(a-2)x^2 - 6ax + \frac{1}{2}a^2$$

$$\text{Solve}[p'[x] == 0, x]$$

$$\{\{x \rightarrow 1\}, \{x \rightarrow -1 - \sqrt{1-a}\}, \{x \rightarrow -1 + \sqrt{1-a}\}\}$$

A1

h. i. Find the values of a for which p has only one stationary point.

1 mark

Answer: $a > 1$.

Solution:

Require only one solution to $p'(x) = 0$.

From **part g.**: $x = 1, -1 \pm \sqrt{1-a}$.

$x = 1$ does not depend on a therefore $x = 1$ must be the only solution.

Therefore the solutions $x = -1 \pm \sqrt{1-a}$ must not exist

therefore $1-a < 0$.

A1

ii. Find the minimum value of p when $a = 2$.

1 mark

Answer: $-\frac{13}{2}$.

Solution:

$$p(x) = \frac{3}{2}x^4 + 2x^3 + 3(a-2)x^2 - 6ax + \frac{1}{2}a^2.$$

$$p(x)|_{a=2} = \frac{3}{2}x^4 + 2x^3 - 12x + 2.$$

For minimum stat point:

$$p'(x) = 0 \quad \Rightarrow \quad x = 1.$$

$$p(1) = \frac{13}{2}.$$

$$p[x_] := \frac{3}{2}x^4 + 2x^3 + 3(a-2)x^2 - 6ax + \frac{1}{2}a^2 /. a \rightarrow 2$$

$p[x]$

$$2 - 12x + 2x^3 + \frac{3x^4}{2}$$

$\text{Solve}[p'[x] == 0, x, \text{Reals}]$

$\{\{x \rightarrow 1\}\}$

$p[1]$

$$-\frac{13}{2}$$

A1.

Note:

The minimum value needs to be stated, not the coordinate of the TP.

- iii. If p has only one stationary point, find the values of a for which $p(x) = 0$ has no real solutions.

2 marks

From **part h.i** there is one stationary point at $x = 1$ when $a > 1$.

The graph of $p(x)$ (when $a > 1$) is a 'happy' quartic (looks like a parabola).

Substitute $x = 1$ into $p(x)$ to get coordinate of TP:

$$p(1) = \frac{1}{2}(a^2 - 6a - 5).$$

Coordinates of turning point:

$$\left(1, \frac{1}{2}(a^2 - 6a - 5)\right).$$

For no x -intercepts (i.e. no real solutions) the y -value of the TP sits above the x -axis:

$$\frac{1}{2}(a^2 - 6a - 5) > 0$$

$$\text{therefore } a \in (-\infty, 3 - \sqrt{14}) \cup (3 + \sqrt{14}, \infty).$$

$$\text{But } a > 1 \text{ therefore reject } a \in (-\infty, 3 - \sqrt{14}).$$

$$\text{Answer: } a \in (3 + \sqrt{14}, \infty).$$

$$p[x_] := \frac{3}{2}x^4 + 2x^3 + 3(a-2)x^2 - 6ax + \frac{1}{2}a^2$$

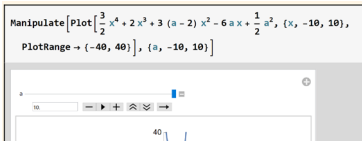
p[1] // Simplify

$$\frac{1}{2}(-5 - 6a + a^2)$$

Reduce[$\frac{1}{2}(-5 - 6a + a^2) > 0, a]$

$$a < 3 - \sqrt{14} \mid \mid a > 3 + \sqrt{14}$$

Use Manipulate to help visualise the question.



M1 $p(1) > 0$

A1:

Correct answer.

Must state rejected interval **with reason**.

END OF QUESTION AND ANSWER BOOK